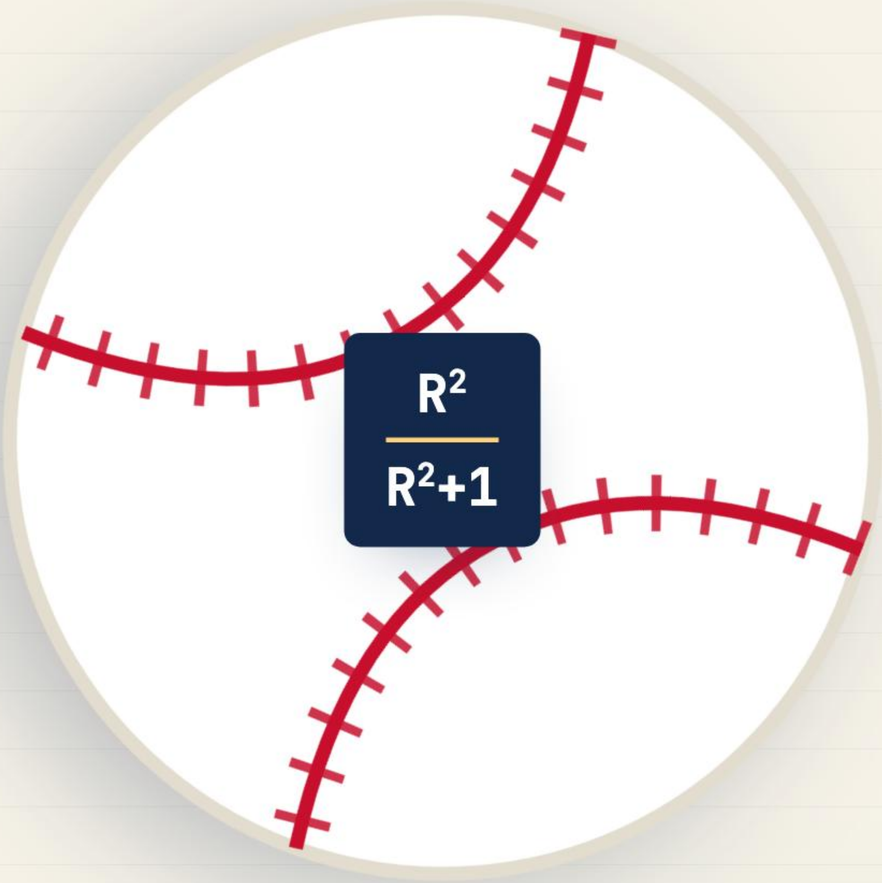
SPM 261 · WEEK 10-1

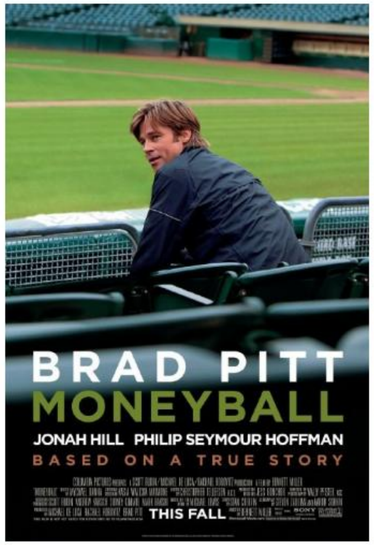
BASEBALL ANALYTICS:

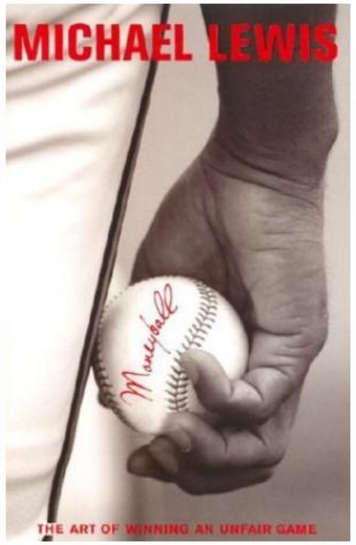
MATHLETICS CH.1

SPM261 Intro to Sport Management



Click, scroll wheel, or press → to continue · N for notes







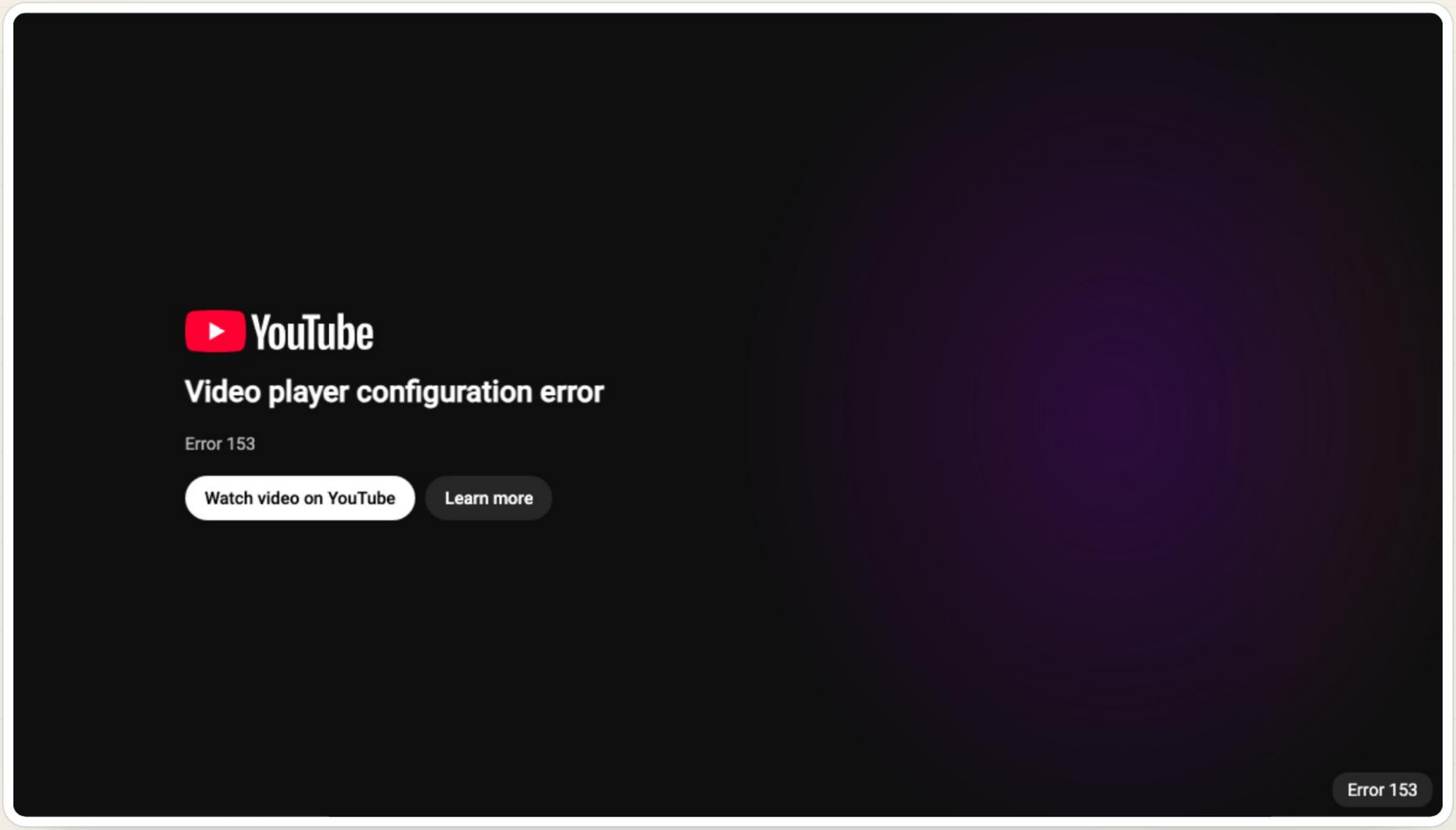
SABERMETRICS

http://m.mlb.com/glossary/advanced-stats

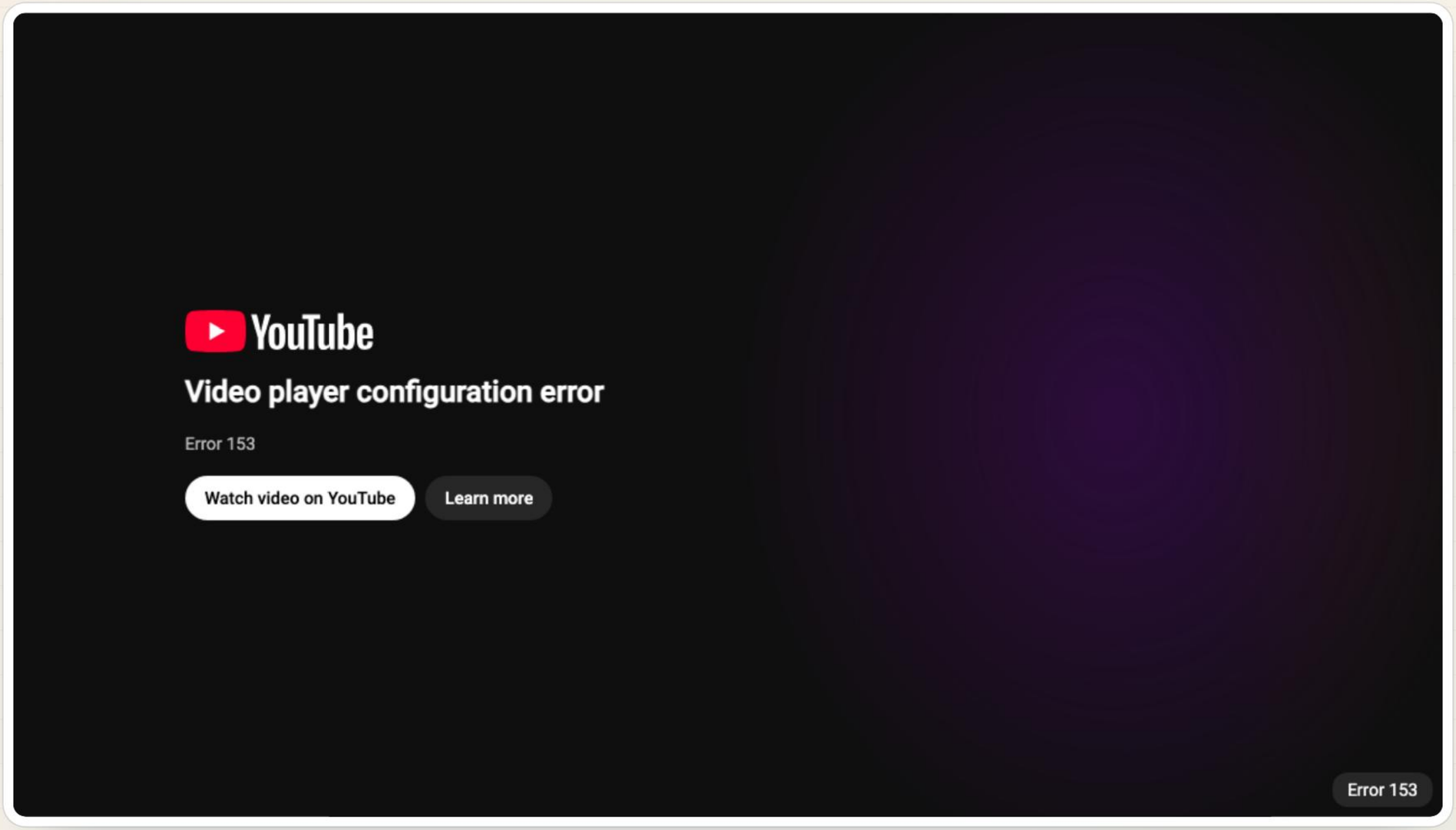








https://www.youtube.com/watch?v=JXyisZLObHc&list=PLZbXA4lyCtqosSLgdOSwUWDK_KH9c2rIq



https://www.youtube.com/watch?v=aNDj-H1jxV0&list=PLZbXA4lyCtqosSLgdOSwUWDK_KH9c2rIq&index=2



WHAT IS SPORT ANALYTICS?



WHAT IS SPORT ANALYTICS?

01 **Data**



WHAT IS SPORT ANALYTICS?

01 Data

02 Analyzing skill



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WHAT IS SPORT ANALYTICS?

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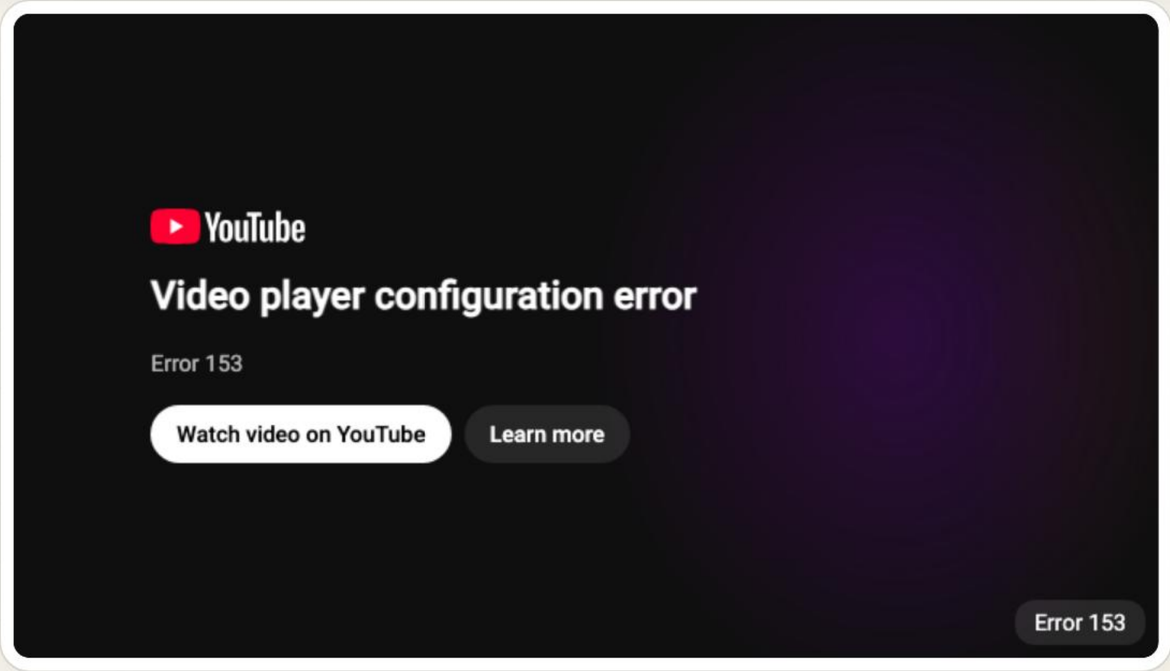
04 Basic features of the data

05 Data visualization

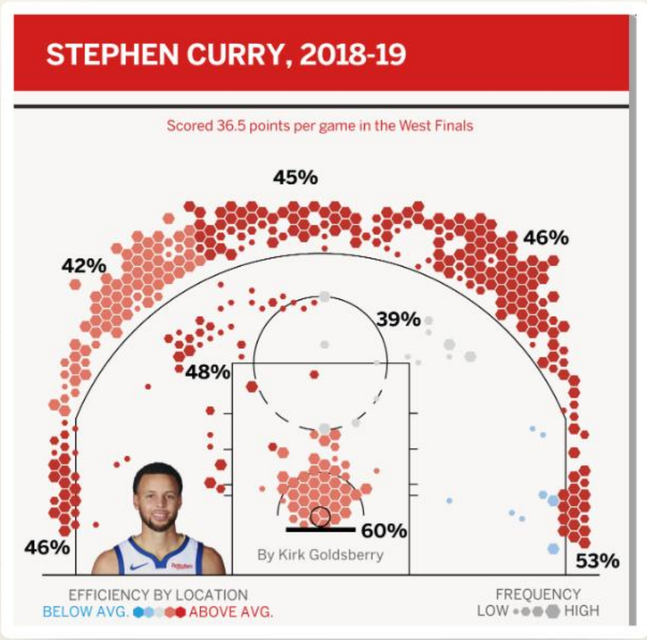
06 Prediction-significance test

WHAT IS SPORT ANALYTICS? CONT.

Player/team performance, winning/ performance prediction, fan emotions/reactions (i.e., social media), economic impact, betting market, rule/ policy test, playoff team selection, etc.



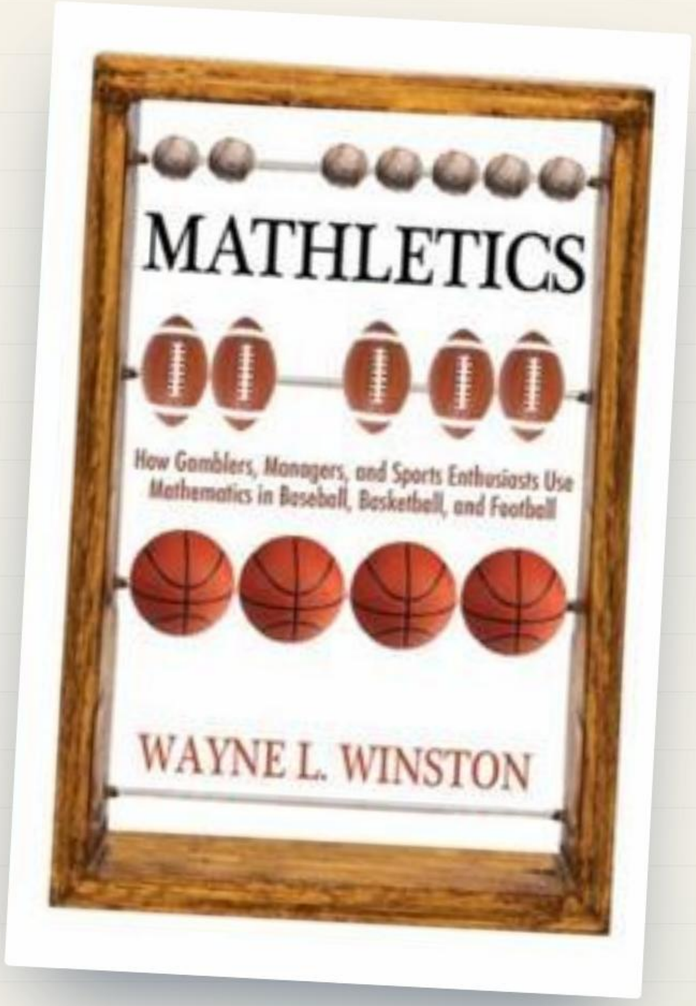
https://www.youtube.com/watch?v=MpLHMKTolVw



PLEASE READ THE BOOK MATHLETICS:

(a) Ch.1 Baseball's Pythagorean Theorem

You can download the book Mathletics on course website. See Module 9





MATHLETICS · CHAPTER 1

BASEBALL'S PYTHAGOREAN THEOREM (CH.1)



PREDICTING A TEAM'S WINNING PERCENTAGE & # OF GAMES

BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

- The more runs a baseball team scores, the more games the team should win.
- Bill James introduces a formula that measures percentage of games won by baseball team (p.3).
- Pythagorean Theorem states that a triangle is a right triangle if and only if $a^2+b^2=c^2$.
- For example, a triangle with sides of lengths 3, 4, and 5 is right triangle because $3^2+4^2=5^2$.

$$\frac{\text{runs scored}^2}{\text{runs scored}^2 + \text{runs allowed}^2} = \text{estimate of percentage of games won.} \qquad (1)$$

This formula has several desirable properties.

- The predicted win percentage is always between 0 and 1.
- An increase in runs scored increases predicted win percentage.
- A decrease in runs allowed increases predicted win percentage.